Calibration of Probabilities

Statistical Methods for Machine Learning — Assignment 2

Salman Askarov
Matriculation number 65829A
Università degli Studi di Milano, A.Y. 2025/26

October 29, 2026

Abstract

A classifier can be accurate while its probabilities are poor estimates of $\eta(x) = \Pr(Y = 1 \mid X = x)$. Following Niculescu-Mizil and Caruana, I measure this for logistic regression (LR) and random forests (RF) on Spambase and Adult, and correct it with Platt scaling and isotonic regression fitted on a held-out calibration set, all implemented from scratch. Over ten random splits, both calibrators lower the Brier score and expected calibration error (ECE) of RF on every split (ECE from 0.053 to 0.020 and 0.017 on Spambase, from 0.024 to 0.012 and 0.010 on Adult) without changing accuracy or AUC, while LR needs little correction, as in the reference study. Isotonic regression has the lowest ECE but a worse log-loss than Platt scaling, because a few of its outputs are wrong and exactly 0 or 1. For the same reason it does not overtake Platt scaling on Adult even with 5,000 calibration examples. Calibrating RF on its training set is worse than not calibrating.

1 Introduction

In binary classification, a probabilistic predictor is meant to estimate $\eta(x) = \Pr(Y = 1 \mid X = x)$, the function that defines the Bayes optimal classifier [4, Statistical Learning; Logistic Regression]. Accuracy only checks whether $\hat{p}(x)$ exceeds $\frac{1}{2}$, and ROC AUC only the order of the predictions, so a model can be accurate but *miscalibrated*: of the cases where it says 0.9, perhaps only 70% are positive. As Niculescu-Mizil and Caruana [7] put it, accuracy and AUC are not sufficient when the probabilities themselves are used. An example is the spam loss of [4, Introduction], which charges 2 for a legitimate e-mail flagged as spam and 1 for the opposite mistake: its optimal rule flags an e-mail when $\eta(x) > \frac{2}{3}$, which requires $\hat{p} \approx \eta$.

Following Niculescu-Mizil and Caruana [7], I train LR and RF, correct them with Platt scaling [9] and isotonic regression [10] fitted on held-out data, and evaluate them on two datasets over ten random splits. Two extra experiments vary the size of the calibration set and fit the calibrators on the training set instead.

2 Background

Calibration. Examples are independent draws from an unknown distribution, and the Bayes optimal classifier for the zero-one loss predicts 1 if and only if $\eta(x) \geq \frac{1}{2}$ [4, Statistical Learning], so thresholding $\hat{p}$ at $\frac{1}{2}$ is justified only if $\hat{p}$ estimates η. Since η is unknown, one checks instead the weaker property of *calibration* [5],

$$\Pr\big(Y = 1 \mid \hat{p}(X) = q\big) = q \qquad \text{for all } q \in [0, 1], \tag{1}$$

which $\hat{p} = \eta$ implies but which the constant base rate also satisfies. A *reliability diagram* plots, for K bins of predictions, the fraction of positives $\bar{y}_k$ against the mean prediction $\bar{p}_k$; a calibrated model lies on the diagonal. The expected calibration error is the weighted distance from it,

$$\mathrm{ECE} = \sum_{k=1}^{K} \frac{n_k}{n} \left|\bar{y}_k - \bar{p}_k\right|, \qquad K = 10, \tag{2}$$

with n_k predictions in bin k. It depends on the binning.

Losses for probabilities. The logarithmic and quadratic losses of [4, Introduction] give, on a test set,

$$\mathrm{LL} = -\frac{1}{n}\sum_{i=1}^{n}[y_i \log \tilde{p}_i + (1-y_i)\log(1-\tilde{p}_i)], \qquad \mathrm{BS} = \frac{1}{n}\sum_{i=1}^{n}(p_i - y_i)^2, \tag{3}$$
$$\tilde{p}_i = \min\big(\max(p_i, 10^{-15}),\, 1-10^{-15}\big).$$

Both are minimized in expectation by η. For the quadratic loss the Bayes optimal prediction is $\mathbb{E}[Y \mid X = x] = \eta(x)$ [4, Statistical Learning]. For the logarithmic loss of $\sigma(g(x))$, i.e. the logistic loss of g, the Bayes optimal g is the log-odds $\ln \frac{\eta}{1-\eta}$, so $\sigma(g^*) = \eta$ [4, Logistic Regression]. Both thus reward calibration and separation together, but they differ in range. The quadratic loss lies in $[0, 1]$, so the Chernoff–Hoeffding bound controls its test estimate [4, Statistical Learning]. The logarithmic loss is unbounded, which discourages predictions too close to 0 or 1 [4, Introduction]: after clipping, a wrong prediction of exactly 0 or 1 costs $-\log 10^{-15} \approx 34.5$, about a hundred times our models' average log-loss. For forecasts with finitely many values p_k, the Brier score splits exactly as

$$\mathrm{BS} = \underbrace{\tfrac{1}{n}\textstyle\sum_k n_k(p_k - \bar{y}_k)^2}_{\text{reliability}} - \underbrace{\tfrac{1}{n}\textstyle\sum_k n_k(\bar{y}_k - \bar{y})^2}_{\text{resolution}} + \underbrace{\bar{y}(1-\bar{y})}_{\text{uncertainty}}, \tag{4}$$

where $\bar{y}$ is the overall fraction of positives. Recalibration targets the reliability term, but a map that merges distinct forecasts also loses resolution.

Why the two models differ. Logistic regression predicts $\sigma(w^\top x)$ by minimizing the logistic loss plus $\frac{\lambda}{2}\|w\|^2$ against overfitting [4, Logistic Regression]. Since the optimum of this loss is the log-odds, LR estimates η well when the log-odds are nearly linear in x (small approximation error) and λ is small, and Niculescu-Mizil and Caruana [7] find it well calibrated on most problems. A tree routes x to a leaf ℓ, and here outputs the leaf's fraction of positives N_ℓ^+/N_ℓ rather than its majority label. Grown to pure leaves, a tree overfits [4, Tree Predictors]. A random forest [3] averages $T = 500$ trees grown on bootstrap samples with random feature subsets, $\hat{p}_{\mathrm{RF}}(x) = \frac{1}{T}\sum_t \hat{p}_t(x)$. Niculescu-Mizil and Caruana [7] explain why the average avoids 0 and 1: near the ends of $[0, 1]$ the trees' variance produces one-sided errors, since the forest predicts 0 only if every tree does. Feature subsetting makes this effect strongest for RF. The result is an S-shaped reliability curve, below the diagonal at low predictions and above it at high ones, which the paper observes for RF on several of its problems.

Platt scaling and isotonic regression. Both map a real score $s(x)$ to a probability by a non-decreasing function fitted on held-out pairs (s_i, y_i). Platt scaling [9] is logistic regression with the single feature s,

$$\hat{p} = \sigma(as + b), \qquad \sigma(z) = 1/(1+e^{-z}), \tag{5}$$

fitted by maximum likelihood ($A = -a$, $B = -b$ in Platt's notation); $a > 1$ pushes probabilities away from $\frac{1}{2}$ and $a < 1$ pulls them towards it. Isotonic regression [10] fits the best non-decreasing step function in squared error, computed by Pool Adjacent Violators (PAV, [1]):

$$\hat{m} = \operatorname*{arg\,min}_{m \text{ non-decreasing}} \sum_{i=1}^{n} w_i\big(y_i - m(s_i)\big)^2. \tag{6}$$

In the decomposition of the risk into estimation error, approximation error and Bayes risk [4, Statistical Learning], Platt scaling has a small estimation error (two parameters) but a nonzero approximation error unless the distortion is a sigmoid in s. Isotonic regression is nonparametric [4, Consistency]: its approximation error vanishes as the calibration set grows, but on small sets its estimation error is large. Accordingly, Niculescu-Mizil and Caruana [7] find Platt scaling better for sigmoid-shaped distortions and small sets, and isotonic regression more powerful with enough data. Isotonic regression's end steps are exactly 0 or 1 when the extreme calibration points share a label, and on its fitting set each step equals its points' mean label, so the reliability term of (4) vanishes there. Both maps are monotone: Platt scaling preserves the AUC, isotonic regression changes it only through ties, and accuracy changes only where the probability crosses $\frac{1}{2}$.

Table 1: Datasets and splits. *Dropped*: rows with a missing value. *Features*: numeric + categorical. *Encoded*: columns after one-hot encoding (for Adult, 103 on seeds 3 and 7, where a category seen in a single row is not in the training set). The split sizes are the same for all seeds. Positive class: spam for Spambase, income `>50K` for Adult.

Dataset	Raw rows	Dropped	Rows	Features	Encoded	Positive	Train	Calibration	Test
Spambase	4,601	0	4,601	57 + 0	57	39.4%	2,761	919	921
Adult	48,842	3,620	45,222	6 + 8	104	24.8%	27,133	9,044	9,045

Why a held-out calibration set. A test error estimates the risk without bias only if the predictor does not depend on the test data, and the training error of an overfitting predictor underestimates its risk [4, Statistical Learning]. The same applies to the data a calibrator is fitted on. Reusing the training set would "introduce unwanted bias": for a model that separates its training set, the sigmoid becomes "just a 0,1 function" [7]. The calibration set therefore plays the role of the development set of [4, Hyperparameter Tuning]: data withheld from training to fit a few extra parameters.

3 Datasets and protocol

Datasets. Both come from OpenML (Table 1). **Spambase** [6] has 4,601 e-mails with 57 numeric, skewed and mostly-zero features (word and character frequencies, capital-letter runs), 39.4% spam, and 394 rows that duplicate another row. **Adult** [2] is a census extract with 6 numeric and 8 categorical features, and the task is to predict an income above \$50K. I drop the 3,620 rows with a missing `workclass` (2,799), `occupation` (2,809) or `native-country` (857), leaving 45,222 rows with 24.8% positives. The dropped rows are only 13.2% positive, a slight population shift. Adult is also one of the paper's problems, with the same 104 encoded attributes, but there models are trained on 4,000 examples and calibrated on 1,000.

Splits, preprocessing and models. Each dataset is split 60/20/20 into training, calibration and test sets, stratified by class (each class is shuffled by a seeded generator and cut at 60% and 80%). Everything is repeated for seeds 0–9, which fix the split, the folds and the forest, and I report mean ± sample standard deviation. As suggested in [4, Introduction], numeric columns are rescaled (standardized with training statistics) and categorical ones binary-encoded (one-hot with the training categories; an unseen category becomes all zeros). The models come from scikit-learn [8]. LR has an L2 penalty of inverse strength $C \in \{10^{-3}, \dots, 10^{2}\}$, and RF has 500 trees, $\sqrt{d}$ features per split and `min_samples_leaf` $\in \{1, 3, 10\}$. The hyperparameter is tuned by 5-fold stratified cross-validation on the training set only [4, Hyperparameter Tuning], scored by log-loss. LR chose $C = 1$ on all Spambase seeds and $C \in \{0.1, 1\}$ (five seeds each) on Adult. RF chose `min_samples_leaf` $= 1$ (pure leaves) on 7 Spambase seeds, and 3 on the other three and on all Adult seeds.

Scores, leakage and metrics. The calibrators take the logit $w^\top x + c$ for LR, and for RF the logit of the clipped vote,

$$s_{\mathrm{RF}}(x) = \mathrm{logit}\big(\min(\max(\hat{p}_{\mathrm{RF}}(x), \delta),\, 1-\delta)\big), \qquad \mathrm{logit}(q) = \log \tfrac{q}{1-q}, \quad \delta = 10^{-6}. \tag{7}$$

On the log-odds scale of η [4, Logistic Regression], the identity is the case $a = 1$, $b = 0$ of (5), so a calibrated model can be left unchanged and $a > 1$ is the stretch RF needs. The clip keeps unanimous votes finite (on Spambase, 95–117 test points on the seeds with `min_samples_leaf` $= 1$, 0–4 otherwise). Isotonic regression depends only on the order of the scores. The training set is used for preprocessing, fitting and tuning, the calibration set only for the calibrators, and the test set only for the final metrics: accuracy (threshold $\frac{1}{2}$), log-loss, Brier score, ECE (10 equal-width bins, as in the paper) and ROC AUC. The *uncalibrated* method is the model's own `predict_proba`. I also count test predictions that are exactly 0 or 1, and the wrong ones among them.

Table 2: Test metrics, mean ± sample standard deviation over 10 seeds. Lower is better for log-loss, Brier score and ECE; higher is better for accuracy and ROC AUC.

Dataset	Model	Method	Accuracy	Log-loss	Brier score	ECE	ROC AUC
Spambase	LR	Uncalibrated	0.9205 ± 0.0067	0.2485 ± 0.0232	0.0639 ± 0.0041	0.0349 ± 0.0051	0.9674 ± 0.0045
Spambase	LR	Platt	0.9178 ± 0.0074	0.2481 ± 0.0178	0.0660 ± 0.0033	0.0413 ± 0.0085	0.9674 ± 0.0045
Spambase	LR	Isotonic	0.9213 ± 0.0066	0.3006 ± 0.0733	0.0623 ± 0.0042	0.0227 ± 0.0047	0.9660 ± 0.0049
Spambase	RF	Uncalibrated	0.9455 ± 0.0081	0.1726 ± 0.0143	0.0451 ± 0.0040	0.0533 ± 0.0101	0.9832 ± 0.0044
Spambase	RF	Platt	0.9457 ± 0.0084	0.1544 ± 0.0199	0.0407 ± 0.0042	0.0195 ± 0.0058	0.9832 ± 0.0044
Spambase	RF	Isotonic	0.9458 ± 0.0089	0.2035 ± 0.0778	0.0412 ± 0.0045	0.0170 ± 0.0063	0.9823 ± 0.0052
Adult	LR	Uncalibrated	0.8481 ± 0.0033	0.3253 ± 0.0041	0.1041 ± 0.0016	0.0109 ± 0.0031	0.9051 ± 0.0026
Adult	LR	Platt	0.8479 ± 0.0034	0.3253 ± 0.0041	0.1040 ± 0.0015	0.0110 ± 0.0031	0.9051 ± 0.0026
Adult	LR	Isotonic	0.8483 ± 0.0034	0.3292 ± 0.0046	0.1043 ± 0.0014	0.0109 ± 0.0039	0.9046 ± 0.0024
Adult	RF	Uncalibrated	0.8618 ± 0.0039	0.3106 ± 0.0047	0.0979 ± 0.0019	0.0244 ± 0.0043	0.9158 ± 0.0034
Adult	RF	Platt	0.8619 ± 0.0046	0.3064 ± 0.0056	0.0970 ± 0.0022	0.0119 ± 0.0041	0.9158 ± 0.0034
Adult	RF	Isotonic	0.8609 ± 0.0043	0.3131 ± 0.0120	0.0971 ± 0.0021	0.0103 ± 0.0028	0.9153 ± 0.0035

4 Implementation

Apart from the base models and their grid search, everything is written in numpy (`src/calib/`).

Platt scaling by Newton's method. As in [9], the labels are replaced by smoothed targets, the method's built-in control against overfitting [7], which keep (a, b) finite even for separable data. The objective is the logistic loss of [4, Logistic Regression] with these soft labels, computed with `logaddexp` so that it never overflows:

$$t_i = \begin{cases} \frac{N_+ + 1}{N_+ + 2}, & y_i = 1, \\ \frac{1}{N_- + 2}, & y_i = 0, \end{cases} \qquad F(a,b) = \sum_{i=1}^{n} \big[\log(1 + e^{z_i}) - t_i z_i\big], \quad z_i = a s_i + b. \tag{8}$$

F is convex in $\theta = (a, b)$. With $x_i = (s_i, 1)^\top$ and $p_i = \sigma(z_i)$,

$$g = \sum_i (p_i - t_i)\, x_i, \qquad H = \sum_i p_i(1 - p_i)\, x_i x_i^\top + 10^{-12} I, \qquad \theta \leftarrow \theta - \eta\, H^{-1} g, \tag{9}$$

starting from $a = 0$, $b = \log \frac{N_+ + 1}{N_- + 1}$. The step length η is halved until F does not increase, and the loop stops when the step norm falls below 10^{-10}. Every run converged in 8 to 11 iterations.

Isotonic regression by PAV. The pairs are stable-sorted by score, and tied scores are merged (weight = count, value = mean label). PAV pushes each point onto a stack of blocks and, while the top block's mean label is not larger than the one below, merges the two. Each point is pushed and popped at most once, so the fit is $O(n)$ after the sort. A new score receives the value of the last block starting at or before it (`searchsorted`); scores below the first block receive the first value.

Metrics and verification. The metrics implement (2)–(3), and the AUC uses the Mann–Whitney rank statistic with average ranks for ties. 78 tests (`tests/`, the only place where scikit-learn's calibration and metric modules are imported) check the metrics against `sklearn.metrics` to 10^{-10}, PAV against `sklearn.isotonic` on data with ties, and Platt scaling without smoothing against an unregularized one-feature logistic regression to 10^{-5}. All seeds are explicit and the forest predicts single-threaded, so `python run_all.py` reproduces every result exactly.

5 Results

Random forest. RF is the more accurate model (Table 2) and the one that gains most from calibration. Both calibrators lower its Brier score and ECE on all ten seeds of both datasets: the ECE falls from 0.0533 to 0.0195 (Platt) and 0.0170 (isotonic) on Spambase, and from 0.0244 to 0.0119 and

0.0103 on Adult. Platt scaling also lowers the log-loss, from 0.1726 to 0.1544 on Spambase (on 9 of 10 seeds) and from 0.3106 to 0.3064 on Adult (on all 10), while isotonic regression raises the mean log-loss on both. This agrees with Niculescu-Mizil and Caruana [7], where RF is mildly miscalibrated on several problems and helped by calibration.

Logistic regression. On Adult, LR is already calibrated, as in the paper: the three methods agree within a fraction of a standard deviation (ECE 0.0109, 0.0110, 0.0109), and Platt scaling learns nearly the identity ($a = 0.99 \pm 0.04$, $b = -0.02 \pm 0.04$). On Spambase (ECE 0.0349), isotonic regression lowers the ECE to 0.0227 and the Brier score from 0.0639 to 0.0623 on all ten seeds, while Platt scaling worsens both (to 0.0413 and 0.0660, on 8 and 9 seeds). Its $a = 0.84 \pm 0.11$ pulls all probabilities towards $\frac{1}{2}$, the failure the paper reports for well-calibrated models, where the sigmoid "has difficulty fitting the predictions in the tails". LR's remaining miscalibration on Spambase (Figure 1) is not sigmoid-shaped, so Platt scaling's approximation error dominates.

Why isotonic regression has a poor log-loss. Its end blocks often contain one class only, so it outputs exactly 0 or 1 on average for 243.8 (LR) and 283.1 (RF) of the 921 Spambase test points, and for 500.4 and 989.0 of the 9,045 Adult ones. Of these, 2.5, 1.7, 1.1 and 2.2 per seed are wrong, at 34.5 each. For Spambase LR this adds $2.5 \times 34.5/921 = 0.094$ to the log-loss, more than the whole gap of 0.052 to the uncalibrated model. The same holds in the other three cases (penalties 0.064, 0.0042, 0.0084 against gaps 0.031, 0.0039, 0.0025). This is the unboundedness of the logarithmic loss [4, Introduction]. With no bound, no Hoeffding-type guarantee keeps its test estimate stable [4, Statistical Learning]: the number of such mistakes varies from 0 to 7 between seeds, and isotonic log-loss on Spambase has a standard deviation of 0.073 and 0.078. The Brier score (at most 1 per mistake) and the ECE (bin averages) barely register these points.

Ranking and accuracy. Platt scaling leaves the AUC unchanged to within 5×10^{-6}. Isotonic regression lowers it by 0.0005 to 0.0014 on average, because its ties count $\frac{1}{2}$ in the Mann–Whitney statistic. Mean accuracy moves by at most 0.003, most for Platt scaling on Spambase LR ($0.9205 \to 0.9178$), where $a < 1$, $b < 0$ shift the score at which the probability crosses $\frac{1}{2}$.

Reliability diagrams. On Spambase (Figure 1, one seed), uncalibrated RF has the S-shape predicted by the variance argument: below the diagonal under 0.4 and above it for most higher predictions. After calibration the curves cross the diagonal back and forth. The rest is mostly noise, since the bins between 0.2 and 0.8 hold only 15 to 40 test points each. On Adult, with ten times more points, the curves are smooth. LR lies on the diagonal throughout, and RF lies above it over 0.5 (underconfident) until calibration brings it back.

Correction maps and predictions. For RF the Platt sigmoid is steeper than $\sigma(s)$ (Figure 2): $a = 1.53 \pm 0.11$ on Spambase and 1.23 ± 0.03 on Adult, with small intercepts ($b = 0.05 \pm 0.11$, 0.11 ± 0.02). It pushes the forest's predictions away from $\frac{1}{2}$, undoing the shrinkage caused by averaging. The isotonic staircase follows the same trend, with 15.0 (RF) and 17.6 (LR) blocks on average on Spambase against 42.9 and 44.7 on Adult (about 50–60 against 200 calibration points per step), and its first and last steps are exactly 0 and 1 in all four seed-0 maps. As in the paper, calibration extends RF's predictions into the tails (Figure 3; on Adult the top bin, almost empty before, fills up), Platt scaling gives smooth histograms and isotonic regression coarse ones. Those few discrete values are the loss of resolution in (4) and the ties that lower the AUC.

Spambase vs Adult. The calibration-set size (919 vs 9,044) controls the estimation error of isotonic regression: the standard deviation of its log-loss is about three (LR) and five (RF) times the uncalibrated one on Spambase, against 1.1 and 2.6 times on Adult, while Platt scaling's stays close to it on both. Both models are more miscalibrated on Spambase (uncalibrated ECE 0.053 vs 0.024 for RF, 0.035 vs 0.011 for LR). My untested hypotheses are that the easier Spambase task has more cases with η near 0

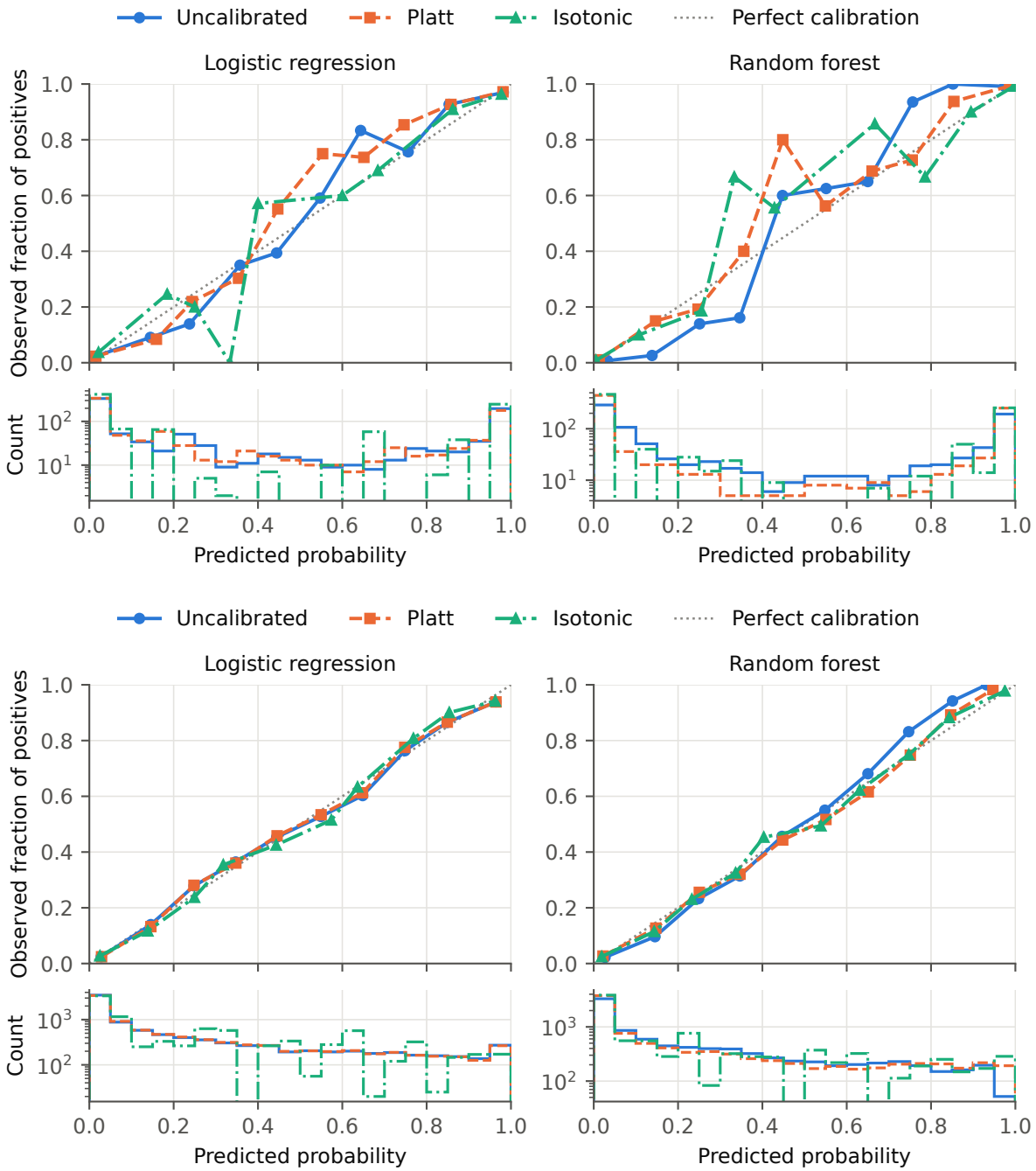


Figure 1: Reliability diagrams on the test set (seed 0), Spambase (top) and Adult (bottom), before and after calibration: fraction of positives against mean predicted probability in 10 equal-width bins (empty bins omitted); dotted: perfect calibration. Below each panel: histogram of the predictions (log scale).

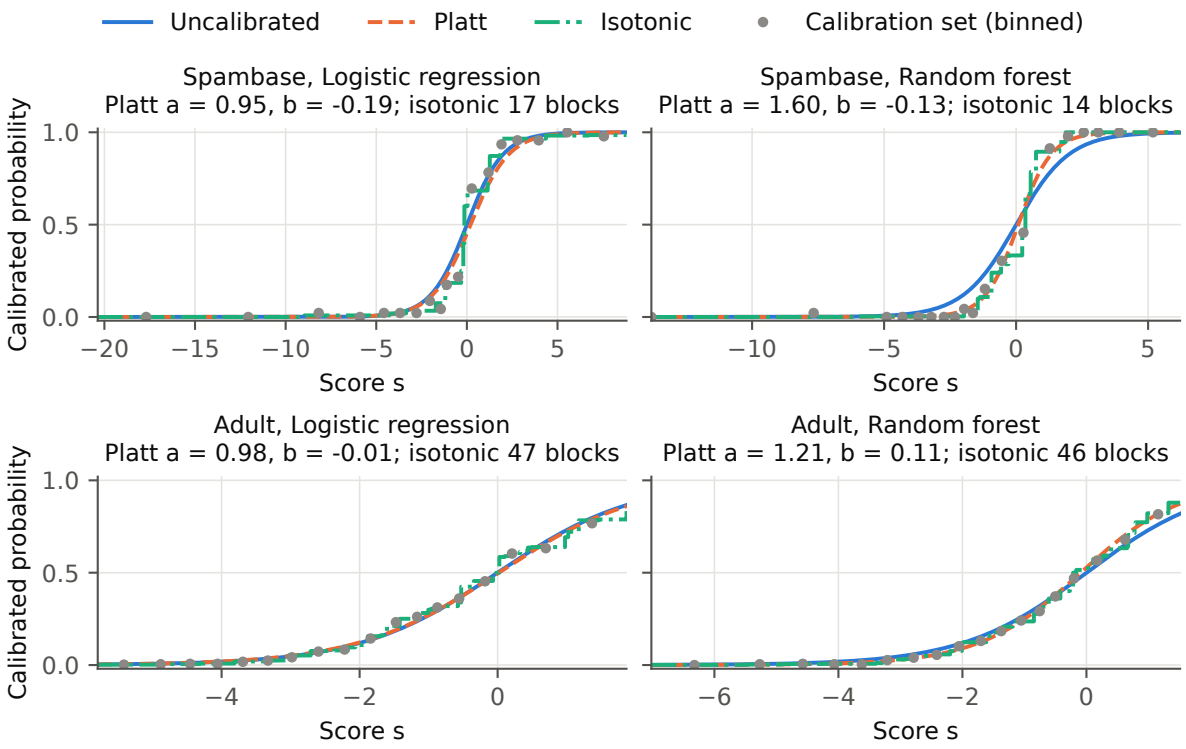


Figure 2: Correction maps learned on the calibration set (seed 0): calibrated probability against score for Platt scaling and isotonic regression, with the uncalibrated map $\sigma(s)$ for reference. Grey dots: fraction of positives of the calibration set in score bins. Horizontal range: 5–95% quantiles of the calibration scores.

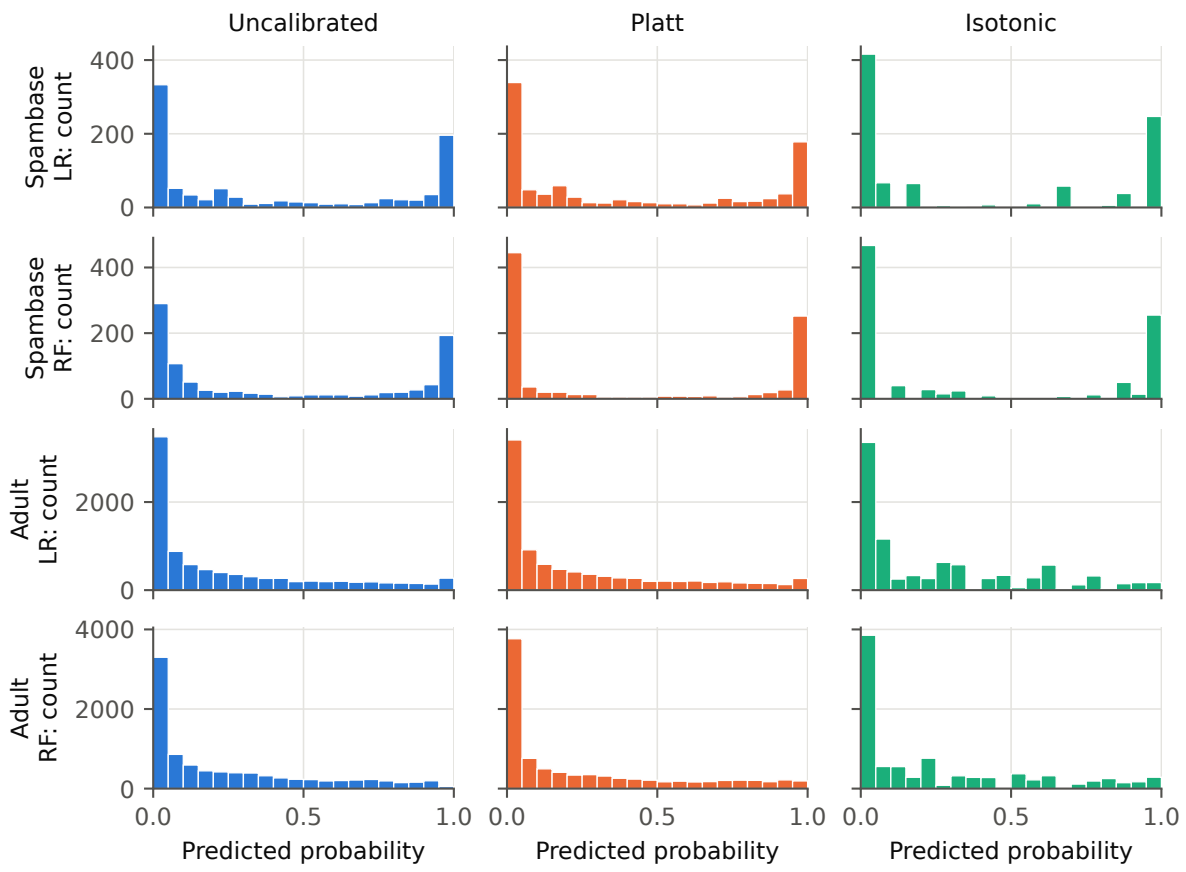


Figure 3: Distribution of the predicted probabilities on the test set (seed 0), before calibration and after Platt scaling and isotonic regression.

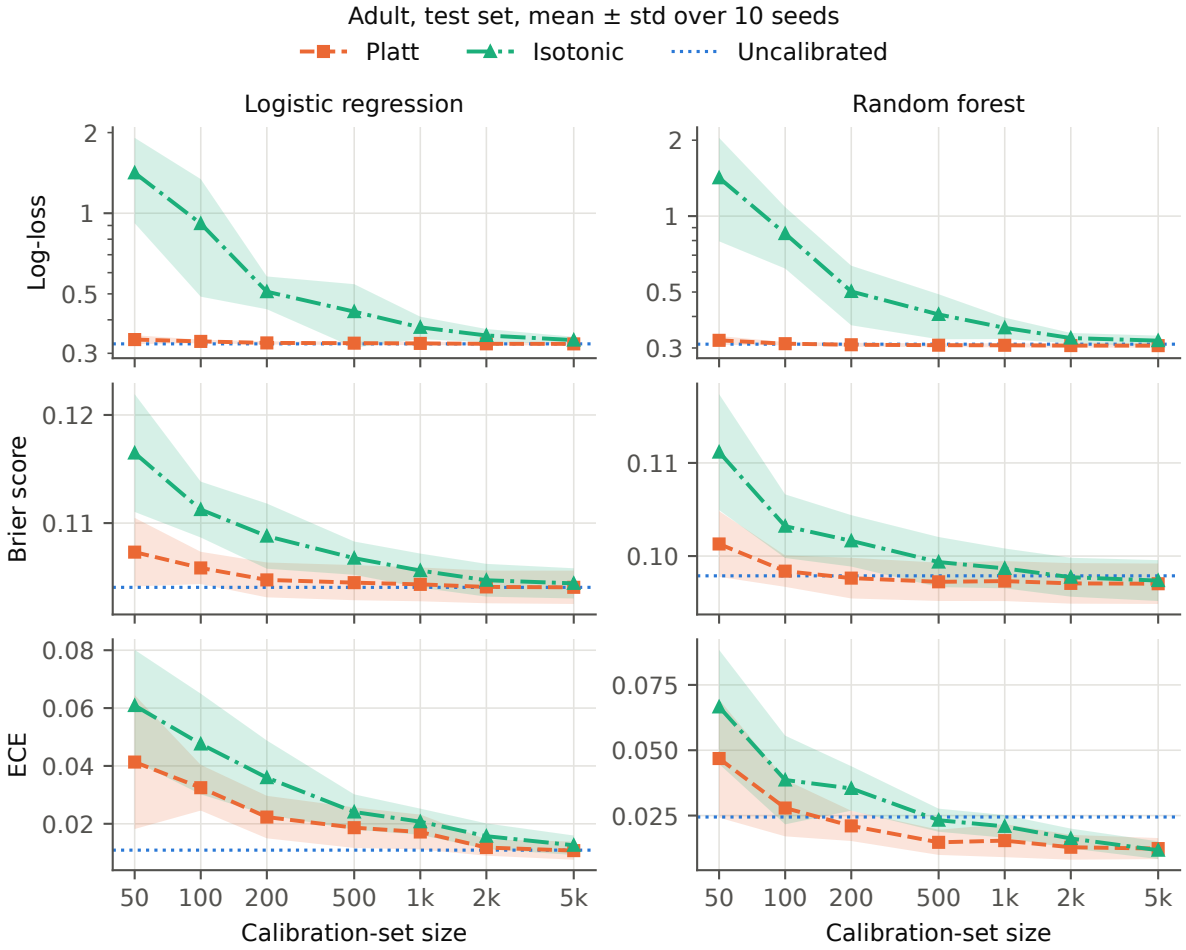


Figure 4: Test log-loss, Brier score and ECE on Adult against the size of the calibration set (stratified subsamples of 50 to 5,000 examples), mean ± one standard deviation over 10 seeds. Dotted: uncalibrated model.

or 1, exactly where the variance argument pulls the forest inwards, and that its log-odds are further from linear in the skewed frequency features, a larger approximation error for LR.

6 Extra experiments

Size of the calibration set. With the Adult base models fixed, the calibrators are refitted on stratified subsamples of 50 to 5,000 examples (Figure 4). Niculescu-Mizil and Caruana [7] report, in squared error averaged over eight problems and nine models, that Platt scaling wins below about 200–1,000 cases and that isotonic regression is as good or better from 1,000. **On Adult the curves do not cross in log-loss or Brier score.** Platt scaling is lower in log-loss on every seed at every size for both models, and in Brier score on at least 9 of 10 seeds, although for RF at 5,000 the Brier gap is small (0.0974 vs 0.0970). Isotonic regression improves fast, as a nonparametric method should (RF log-loss 1.42 at 50 examples, 0.36 at 1,000, 0.32 at 5,000). At 5,000, though, it is still above Platt scaling (0.3065) and the uncalibrated model (0.3106): its 4.4 wrong 0/1 outputs per seed cost about 0.017, more than the gap of 0.014. Only the ECE converges as expected (RF at 5,000: 0.0117 vs 0.0124; at the full 9,044 isotonic regression is ahead, 0.0103 vs 0.0119). I report this as a negative result, not a bug. My hypothesis is that Adult's correction is nearly sigmoid in the logit (Figure 2), so Platt scaling's approximation error is small, the case in which the paper finds the methods "virtually indistinguishable" for large sets. The paper's other learning-curve findings are reproduced. Calibrating an already calibrated model "hurts performance when the calibration sets are small": with 50 or 100 examples, Platt scaling raises LR's ECE from 0.011 to 0.041 and 0.032. RF is helped "once the calibration sets are large enough": about 200 examples for Platt scaling (log-loss and ECE), and 500

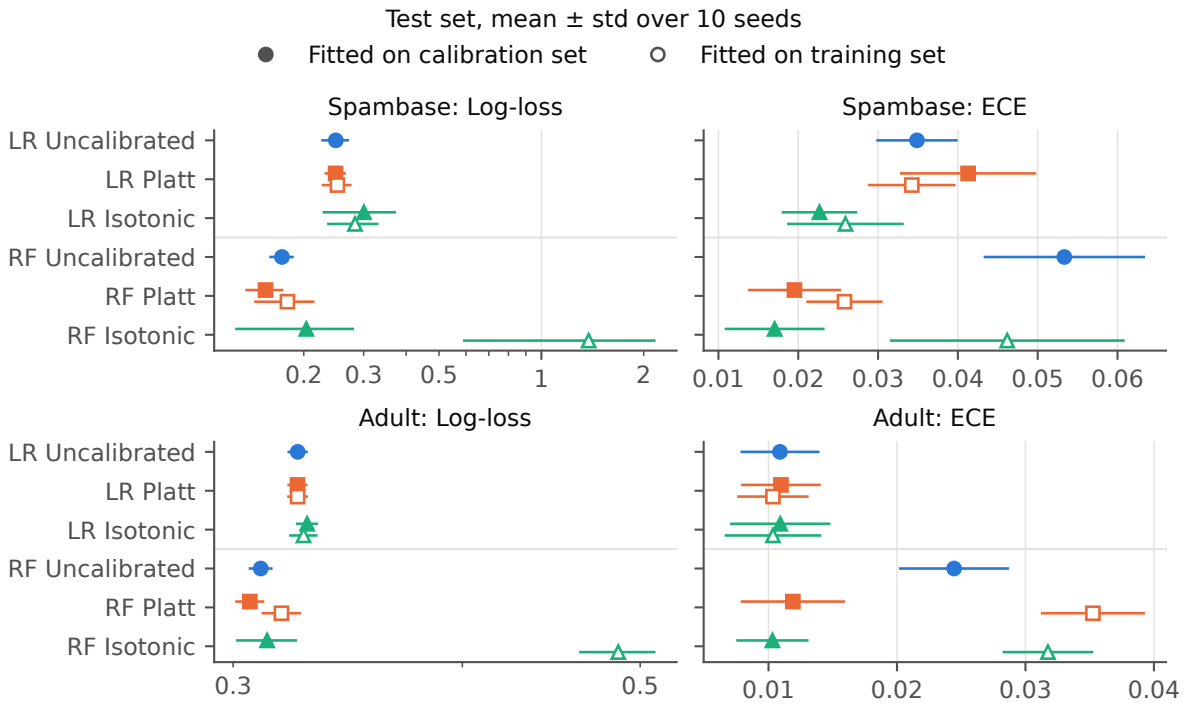


Figure 5: Test log-loss (log scale) and ECE when the calibrators are fitted on the training set (hollow markers) or on the held-out calibration set (filled markers), mean ± one standard deviation over 10 seeds.

(ECE) or 2,000 (Brier score) for isotonic regression.

Calibrating on the training set. Figure 5 tests the held-out argument of Section 2. For RF, fitting on the training set worsens the test log-loss on every seed of both datasets for both calibrators, and on average it is worse than no calibration. Each training point is in the bootstrap sample of about 63% of the trees, and with `min_samples_leaf` $= 1$ (most Spambase seeds) those trees typically route it to a pure leaf of its own label [4, Tree Predictors], so its training vote is optimistic. A vote of 0.7 is almost always right on the training set, and Platt scaling learns a much steeper map ($a = 2.69$ on Spambase and 1.75 on Adult, against 1.53 and 1.23 held out), which is overconfident on test data (Adult ECE 0.0353 against 0.0244 uncalibrated). Isotonic regression produces the paper's "0,1 function" almost literally. On Spambase it keeps 6.3 blocks and outputs exactly 0 or 1 for 809.8 of the 921 test points, and the 35.6 wrong ones alone add $35.6 \times 34.5/921 = 1.34$ to a log-loss of 1.377. For LR the choice matters little: a regularized model with 57 or 104 weights fitted on thousands of examples has a small estimation error, so its training scores look like its test scores. On Spambase, Platt scaling fitted on the three times larger training set even lowers the test Brier score on all ten seeds.

7 Discussion and limitations

The results fit the bias–variance picture of Section 2 and largely reproduce the reference study. RF, an average of high-variance trees, is miscalibrated in the predicted direction and should be recalibrated. Platt scaling on the logit of the vote is a safe default: it improved all three probability metrics, needed only a few hundred examples and never produced degenerate outputs. LR, whose loss is minimized by η itself, should be recalibrated only if its reliability diagram shows a problem, and where that problem is not sigmoid-shaped (Spambase), only the low-bias isotonic map helps. The main departure from the paper is that the metric decides the verdict on isotonic regression: best in ECE, worst in log-loss. Bounding its outputs away from 0 and 1 would likely remove most of the penalty; I did not try it, to keep to the specified method.

Limitations. (i) Two datasets are a small basis compared with the paper's eight. (ii) The calibration set costs 20% of the data; cross-validated calibration would use all of it. (iii) For a loss in $[0, 1]$, the Chernoff–Hoeffding bound [4, Statistical Learning] gives a 95% interval of ± 0.045 on one Spambase test set and ± 0.014 on Adult, wider than most differences found. The evidence is therefore the consistency of the paired comparisons over ten seeds, which is not a formal test, since the splits share data. (iv) The ECE depends on the binning and is noisy in Spambase's sparse middle bins. (v) Dropping Adult's incomplete rows (13.2% positive) restricts the results to the complete-case population, and Spambase's 394 duplicated rows can fall in both training and test. (vi) The RF score clip affects Platt scaling on unanimous votes, and the figures show one seed.

8 Conclusion

With Platt scaling (Newton's method), isotonic regression (PAV), the metrics and the data pipeline implemented from scratch, I reproduced the main qualitative findings of Niculescu-Mizil and Caruana [7] on two datasets. The random forest is underconfident, with the S-shaped reliability curve of averaged high-variance trees, and both calibrators fix most of this without changing its ranking or accuracy. Logistic regression, whose loss is minimized by η, needs little correction, and calibration must use held-out data. Unlike the paper's averages, on Adult Platt scaling beat isotonic regression at every calibration-set size tried, because the correction is nearly sigmoid and isotonic regression's exact 0/1 outputs are heavily penalized by the unbounded logarithmic loss.

Declaration

I declare that this material, which I now submit for assessment, is entirely my own work and has not been taken from the work of others, save and to the extent that such work has been cited and acknowledged within the text of my work. I understand that plagiarism, collusion, and copying are grave and serious offences in the university and accept the penalties that would be imposed should I engage in plagiarism, collusion or copying. This assignment, or any part of it, has not been previously submitted by me or any other person for assessment on this or any other course of study.

Salman Askarov October 29, 2026

References

[1] Miriam Ayer, H. D. Brunk, G. M. Ewing, W. T. Reid, and Edward Silverman. An empirical distribution function for sampling with incomplete information. *The Annals of Mathematical Statistics*, 26(4):641–647, 1955.

[2] Barry Becker and Ronny Kohavi. Adult. UCI Machine Learning Repository, 1996.

[3] Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001.

[4] Nicolò Cesa-Bianchi. Statistical methods for machine learning: Lecture notes. Università degli Studi di Milano, 2025. Versions of 2023–2025.

[5] Morris H. DeGroot and Stephen E. Fienberg. The comparison and evaluation of forecasters. *Journal of the Royal Statistical Society. Series D (The Statistician)*, 32(1/2):12–22, 1983.

[6] Mark Hopkins, Erik Reeber, George Forman, and Jaap Suermondt. Spambase. UCI Machine Learning Repository, 1999.

[7] Alexandru Niculescu-Mizil and Rich Caruana. Predicting good probabilities with supervised learning. In *Proceedings of the 22nd International Conference on Machine Learning (ICML)*, pages 625–632, 2005. doi: 10.1145/1102351.1102430.

[8] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, Mathieu Blondel, Peter Prettenhofer, Ron Weiss, Vincent Dubourg, Jake Vanderplas, Alexandre Passos, David Cournapeau, Matthieu Brucher, Matthieu Perrot, and Édouard Duchesnay. Scikit-learn: Machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.

[9] John C. Platt. Probabilistic outputs for support vector machines and comparisons to regularized likelihood methods. In Alexander J. Smola, Peter Bartlett, Bernhard Schölkopf, and Dale Schuurmans, editors, *Advances in Large Margin Classifiers*, pages 61–74. MIT Press, 1999.

[10] Bianca Zadrozny and Charles Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 694–699, 2002.